

K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

Semester: January –May 2021

In-Semester Examination

(PWD)

Class: SY B. Tech

Branch: Computer

Full name of the course: Probability, Statistics & Optimization Techniques

Duration: 1hr.15 min (attempting questions)

+15 min (uploading)

Semester: IV

Course Code: 2UCC401

Max. Marks: 30

Q. No		Questions	Marks																						
Q1	1.1	From the following probability distribution table, find the constant k	10 marks (2 Mark Each)																						
		<table><tr><td>X</td><td>0</td><td>10</td><td>15</td></tr><tr><td>P(X=x)</td><td>(k-6)/5</td><td>2/k</td><td>14/5k</td></tr></table>		X	0	10	15	P(X=x)	(k-6)/5	2/k	14/5k														
		X		0	10	15																			
	P(X=x)	(k-6)/5		2/k	14/5k																				
	(a) 8 (b) 3 (c) 5 (d) 7																								
	1.2	<table><tr><td></td><td>Y=2</td><td>Y=4</td><td>Y=5</td></tr><tr><td>X=1</td><td>1/12</td><td>1/24</td><td>1/24</td></tr><tr><td>X=2</td><td>1/6</td><td>1/12</td><td>1/8</td></tr><tr><td>X=3</td><td>1/4</td><td>1/8</td><td>1/12</td></tr></table>			Y=2	Y=4	Y=5	X=1	1/12	1/24	1/24	X=2	1/6	1/12	1/8	X=3	1/4	1/8	1/12						
		Y=2	Y=4	Y=5																					
X=1		1/12	1/24	1/24																					
X=2		1/6	1/12	1/8																					
X=3	1/4	1/8	1/12																						
Find conditional prob $P(X \leq 2/Y = 4)$																									
(a) 1/8 (b) 1/4 (c) 1/2 (d) 1/6																									
1.3	Find coefficient of correlation if line of regression of y on x is $10y - 6x = 80$ and line of regression of x on y is $7y - 10x = 2$ (a) 0.42 (b) 0.60 (c) 0.69 (d) - 0.60																								
1.4	The probability that at any moment one telephone line out of 5 will be busy is 0.2. what is the probability that 4 lines are busy? (a) 0.0064 (b) 0.00128 (c) 0.0819 (d) 0.4096																								
1.5	Find 95% confidence interval of the odds ratio for logistic modal is given by $\log\left(\frac{p}{1-p}\right) = 0.69x - 0.24$ if value of z for 95% confidence interval is 1.96 and standard error is 0.11. (a) (0.47, 0.91) (b) (1.61, 2.47) (c) (-0.46, -0.02) (d) (0.63, 0.98)																								
Q2	A	<table><tr><td>X</td><td>100</td><td>110</td><td>120</td><td>130</td><td>140</td><td>150</td><td>160</td><td>170</td><td>180</td><td>190</td></tr><tr><td>Y</td><td>45</td><td>49</td><td>54</td><td>61</td><td>66</td><td>70</td><td>73</td><td>78</td><td>85</td><td>88</td></tr></table>	X	100	110	120	130	140	150	160	170	180	190	Y	45	49	54	61	66	70	73	78	85	88	5 marks (1+1+1+1+1)
		X	100	110	120	130	140	150	160	170	180	190													
		Y	45	49	54	61	66	70	73	78	85	88													
From the above data																									
(i)prepare the table with appropriate columns (ii)Find mean $\bar{x}, \bar{y}$																									

		(iii)find regression coefficient b_{yx} (iv)find regression equation to estimate y (v) estimate y corresponding to x=155																						
B	<table><tr><td>x</td><td>229</td><td>226</td><td>228</td><td>227</td><td>230</td><td>232</td><td>223</td><td>225</td><td>232</td><td>228</td></tr><tr><td>y</td><td>250</td><td>261</td><td>236</td><td>259</td><td>234</td><td>263</td><td>264</td><td>267</td><td>250</td><td>256</td></tr></table> From the above data (i)prepare the table with appropriate columns (ii)write formula to find Karl Pearson’s correlation between x & y. (iii) calculate Karl Pearson’s correlation between x & y.	x	229	226	228	227	230	232	223	225	232	228	y	250	261	236	259	234	263	264	267	250	256	5 marks (2+1+2)
	x	229	226	228	227	230	232	223	225	232	228													
	y	250	261	236	259	234	263	264	267	250	256													
	OR																							
<table><tr><td>x:</td><td>68</td><td>64</td><td>75</td><td>50</td><td>64</td><td>80</td><td>75</td><td>40</td><td>55</td><td>64</td></tr><tr><td>y:</td><td>62</td><td>58</td><td>68</td><td>45</td><td>81</td><td>60</td><td>68</td><td>48</td><td>50</td><td>70</td></tr></table> From the above data (i)prepare the table with appropriate columns (ii)write formula to find coefficient of rank correlation between x & y. (iii) calculate coefficient of rank correlation between x & y.	x:	68	64	75	50	64	80	75	40	55	64	y:	62	58	68	45	81	60	68	48	50	70	5 marks (2+1+2)	
x:	68	64	75	50	64	80	75	40	55	64														
y:	62	58	68	45	81	60	68	48	50	70														
Q3		Attempt any TWO out of THREE	(5+5) marks																					
	A	In a certain college 4% of the boys and 1% of the girls are taller than 1.8 m. Furthermore 40% of the students are girls. (i)write different probabilities from the above data (ii)write formula for Bay’s theorem (iii)if a student is selected at random and taller than 1.8 m what is probability that the student is girl?	5 marks (2+1+2)																					
	B	In a distribution exactly normal 15% of items are under 45 & 79% are under 61. If for this normal distribution, m is the mean & σ is standard deviation (i)write different probabilities from the above data (ii) form two equations (iii)find the value of mean & standard deviation.	5 marks (2+1+2)																					
	C	A continuous random variable X has the probability density function $f(x) = ke^{-x}, x \geq 0$ (i) Find k (ii) Find mean (iii)Find variance	5 marks (2+1+2)																					